Network Analysis of the 15-Minute City Model in Atlanta's West End

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ABSTRACT

The rapid growth of cities presents challenges in creating sustainable, inclusive, and resilient urban environments. Car dependency, uneven service distribution, and limited access to essential services exacerbate urban inequities. The 15-minute city model addresses these issues by promoting walkable neighborhoods where essential services—work, education, grocery, healthcare, and recreation—are accessible within a 15-minute radius. This research evaluates Atlanta's West End neighborhood using network and graph theory to identify gaps in connectivity. Findings suggest that car-centric design limits West End's ability to meet 15-minute city standards. The study identifies specific gaps and proposes improvements to foster sustainable development, with future analysis expanding to the entire city of Atlanta.

INTRODUCTION

Urban planning models like the 15-minute city promote sustainable living by ensuring essential services are accessible within a 15-minute walk or bike ride. Compared to cities like Rome, London, and Paris, which have successfully integrated the 15-minute city concept, Atlanta's West End neighborhood exhibits substantial challenges in connectivity. Atlanta's broader car-centric infrastructure, mixed-use zoning, and gaps in walkability impede West End's alignment with this model.

Known for its rich history and vibrant community, the West End is a transit-accessible area with significant potential for sustainable development. However, the neighborhood's urban design reflects Atlanta's car-centric approach, limiting its alignment with the 15-minute city standards. It features a mix of residential areas, cultural landmarks, and commercial spaces but faces connectivity gaps and inconsistent walkability.

This research applies network and graph theory to evaluate the accessibility and connectivity of essential services in the West End. Metrics like centrality, clustering coefficients, and shortest path analysis are used to assess service accessibility, highlighting areas needing improvement. Insights from this study aim to inform sustainable urban planning practices, bridging existing gaps in Atlanta's infrastructure.

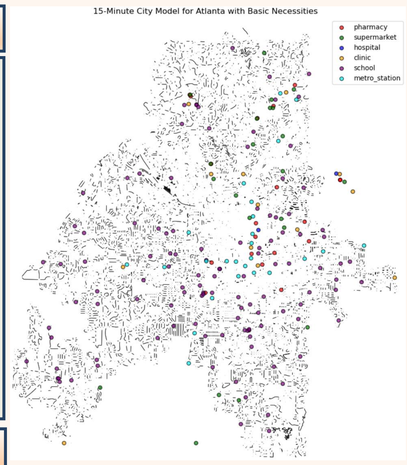
AIMS AND HYPOTHESES

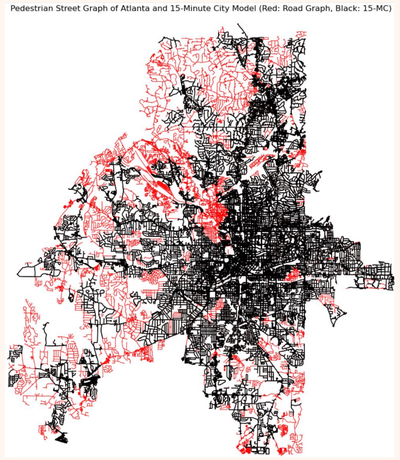
Aims

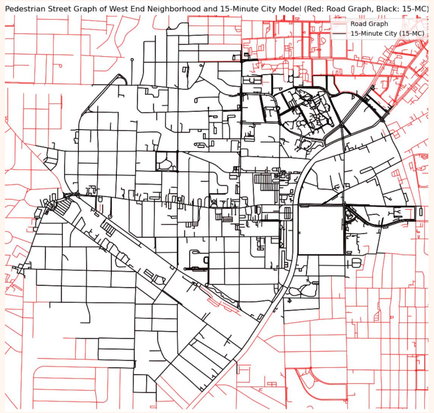
1. Evaluate how well Atlanta's West End aligns with the 15-minute city model.
2. Quantify the accessibility and connectivity of essential services using graph-based metrics.
3. Provide actionable recommendations for achieving the 15-minute city vision in West End.

Hypotheses

- H1: The West End neighborhood demonstrates uneven service distribution and limited pedestrian connectivity, falling short of the 15-minute city standards.
- H2: Centrality and clustering metrics will highlight key areas with high service density and others requiring targeted improvements.
- H3: Proposed infrastructure enhancements will significantly improve the area's alignment with 15-minute city principles.







Area	Total Edges (|E|)	15- Minute City Edges (|C|)	|C|/|E| Ratio
West End Neighborhood	2846	2818	0.99
Atlanta (whole)	180838	6272	0.03

METHODS

1. **Data Sources:**
   - **Geographic Data:** OpenStreetMap (OSM) for road networks and infrastructure mapping.
   - **Amenity Data:** Healthcare facilities, grocery stores, parks, and educational institutions from public databases.
   - **Demographic Data:** Population density, income levels, and car ownership from U.S. Census and local surveys.
   - **Public Transit Data:** Routes and stops from Atlanta's transit authority.
2. **Software and Tools:**
   - **OSMnx:** Python library for retrieving and analyzing urban street networks.
   - **NetworkX:** For graph-based analysis of connectivity and accessibility metrics.
   - **GIS Software:** ArcGIS or QGIS for spatial visualization.
   - **Python Libraries:** Pandas and Matplotlib for data analysis and visualization.
3. **Graph Construction and Analysis:**
   - Import road networks and service locations from OSM using OSMnx.
   - Construct a graph where nodes represent intersections and services, and edges represent pathways.
   - Use **Centrality Metrics** to identify service hubs, **Clustering Coefficients** to assess local connectivity, and **Shortest Path Analysis** to evaluate service accessibility within a 15-minute walk.

RESULTS AND DISCUSSION

•Accessibility and Connectivity Findings

The calculated |C|/|E| ratio reveals a significant contrast between the West End neighborhood and Atlanta as a whole. The West End neighborhood's ratio of 0.99 indicates that nearly all of its pedestrian street network aligns with the 15-minute city model, demonstrating a high level of local accessibility and connectivity. However, Atlanta as a whole shows a much lower |C|/|E| ratio of 0.03. This result suggests a vast majority of Atlanta's pedestrian network falls outside the 15-minute city standards, highlighting uneven service distribution and substantial gaps in connectivity.

•Service Reachability Analysis

In the West End, pharmacies, grocery stores, and schools are predominantly reachable, aligning with the 15-minute model in most areas. However, healthcare facilities and public transportation show significant accessibility gaps, impacting the uniformity of service coverage. The visual heatmap highlights clustered service availability around key intersections, with underserved regions in other parts of the neighborhood.

•Discrepancy in Atlanta-Wide Analysis

Despite the numerical ratio of 0.03, visual analysis of the Atlanta pedestrian street network shows that the areas with service coverage appear more substantial than the ratio suggests. This discrepancy indicates a limitation in the current methodological approach—potentially due to generalization across diverse neighborhoods or the interpretation of the connected subgraph. Further analysis is required to reconcile these differences, suggesting limitations in the uniformity of urban infrastructure and walkability across Atlanta.

DISCUSSION

•Limitations

The discrepancy noted between the calculated |C|/|E| ratio for Atlanta as a whole and the visual analysis reveals a methodological limitation. The generalization of network metrics across neighborhoods of varying characteristics likely contributed to an underestimation of accessible areas. This suggests that a more nuanced analysis is needed, particularly in cities with diverse infrastructure and zoning practices.

•Recommendations for Improvement

1.Increased Connectivity: Enhancing pedestrian pathways, especially near healthcare facilities and transit stops, could improve accessibility across underserved areas.

1.Public Transport Integration: Increasing bus stops and transit options within the West End would address accessibility gaps, making the neighborhood more aligned with the 15-minute city model.

1.Equitable Service Distribution: To ensure equitable access, new clinics, parks, and other amenities should be strategically placed to cover underserved parts of Atlanta, improving clustering and centrality.

CONCLUSIONS

Summary

The findings of this research indicate that, while Atlanta's West End neighborhood aligns well with the 15-minute city model (|C|/|E| ratio = 0.99), the city as a whole does not. The hypothesis that the West End would struggle to meet these standards was proven incorrect. However, the hypothesis regarding Atlanta's overall connectivity challenges was confirmed, with only 3% of the street network meeting the standards of the 15-minute city model. Key barriers include Atlanta's car-centric urban design, uneven service distribution, and significant gaps in walkability across neighborhoods.

Implications

To foster more equitable and sustainable urban development, targeted infrastructure enhancements are necessary. Improved pedestrian pathways, increased service availability, and better public transportation integration will enhance the connectivity and accessibility of neighborhoods like the West End. Addressing these gaps offers an opportunity to use the West End as a pilot model for broader improvements across Atlanta.

FUTURE RESEARCH

While this research focuses on Atlanta's West End neighborhood, expanding the analysis to the entire city offers valuable insights into Atlanta's broader walkability and connectivity. By applying the same graph-based methods citywide, future studies can assess how well Atlanta's infrastructure supports the 15-minute city model across diverse neighborhoods. Such research could reveal systemic patterns in accessibility gaps, informing targeted citywide improvements. Additionally, integrating demographic data more deeply could provide a nuanced understanding of equity in access to essential services, ensuring that interventions promote inclusive urban development.

Additionally, methodological adjustments are required to address discrepancies between visual representations and calculated ratios in city-wide evaluations, to better understand the variability in walkability and accessibility.

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